

Channel capacity and an introduction to information theory

440-2216/01 Kvalita služeb • Lecture 01

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Where we are going

Part 1

Signals and PCM

What carries information, and how an analog signal becomes bits: PCM, sampling, aliasing

↔ Exercise 01, Step 1

Parts 2–4

Channel capacity

Gaussian noise, SNR, symbols vs bits, and the Shannon–Hartley bound

↔ Exercise 01, Steps 2–4

Part 5

Information

Hartley's measure, Shannon entropy, and how a source model guides compression

↔ README questions

Two questions run through the whole lecture: **how many bits can a channel carry**, and **how many bits does a message actually need**.

What you should be able to do

1. Compute a PCM source rate from sampling frequency and bits per sample.
2. Convert SNR from decibels and calculate an AWGN capacity bound.
3. Distinguish source rate, raw coded bit rate, and delivered payload rate.
4. Calculate entropy and average code length for a small discrete source.

Keep asking: **which signal, which channel, which rate, and which assumptions?**

Channel conditions set a capacity limit

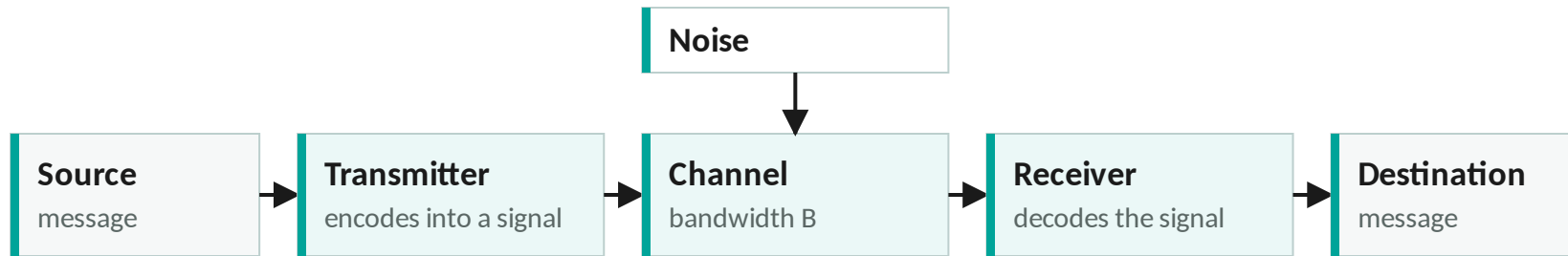
QoS manages rate, delay, jitter, and loss within available resources

Part 1

Signals and PCM — from waveform to bits



A signal is a physical quantity that carries information



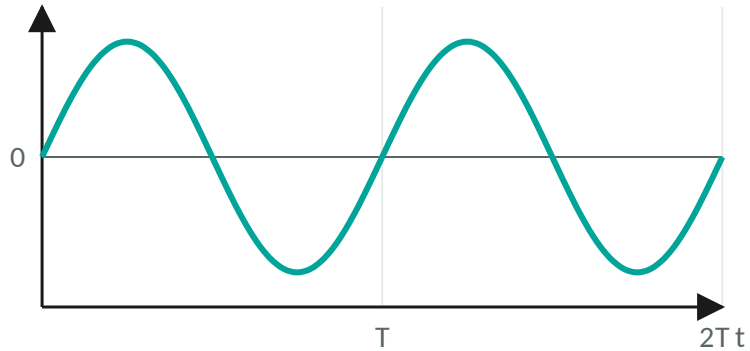
Telephony: microphone → network → earpiece. Information theory: message → channel symbols → message.

In telecommunications — a voltage, a current, an electromagnetic wave that varies in time.

In information theory — a sequence of input symbols or a waveform used to encode a message.

Continuous time, discrete time

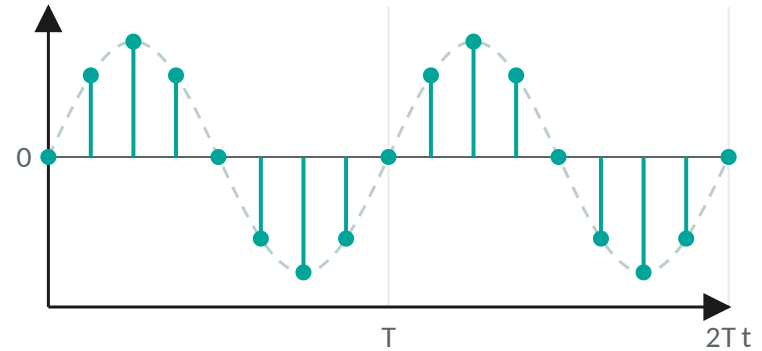
CONTINUOUS TIME



period T · frequency $f = 1/T$

Period T [s] — duration of one repetition. Frequency $f = 1/T$ [Hz] — repetitions per second.

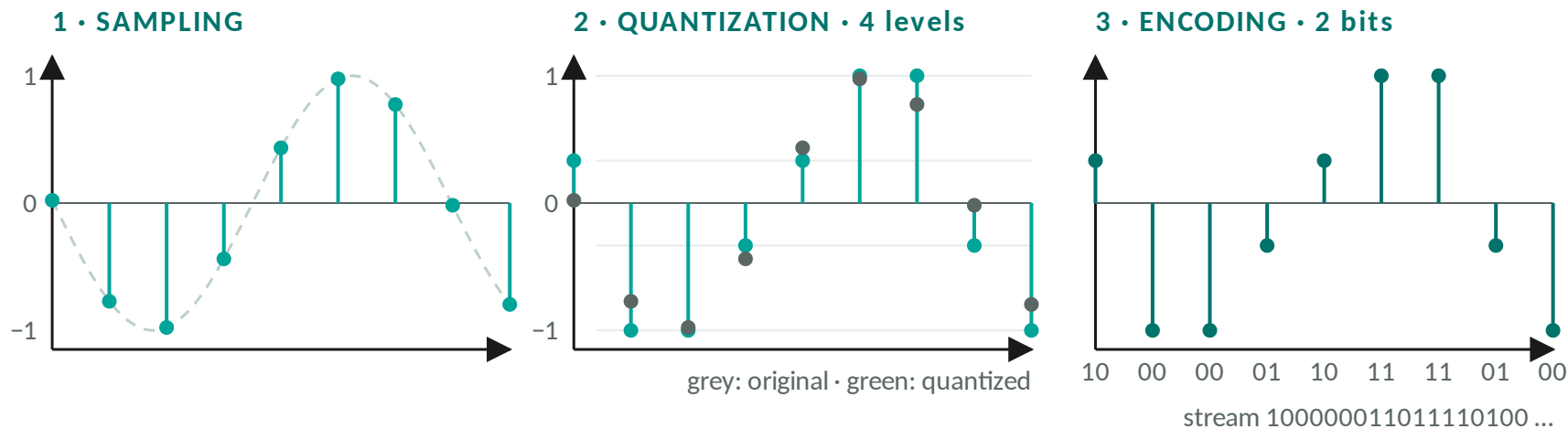
DISCRETE TIME · SAMPLED



sampling period $T_s = 1 / f_s$

Sampling period T_s [s] — spacing of the samples. Sampling frequency $f_s = 1/T_s$ [Hz] — samples per second.

PCM in three steps



b bits per sample $\rightarrow 2^b$ levels • G.711 telephony: 8 000 samples/s $\times$ 8 bits = 64 kbit/s

- 1. Sampling** — read the signal every T_s . Discrete in time, still any amplitude.
- 2. Quantization** — round each sample to the nearest of a finite set of levels. The rounding error is *quantization noise*.
- 3. Encoding** — give each level a binary code word. b bits per sample $\rightarrow$ up to 2^b levels.

The sampling theorem

$$f_s > 2 f_{\max}$$

For an ideally low-pass band-limited signal, exact reconstruction from **ideal, unquantized samples** is possible when the sampling frequency exceeds **twice the highest frequency present**.

Components above $f_s/2$ fold into the sampled baseband: **aliasing**. Ordinary reconstruction cannot tell the overlapping components apart. Use an anti-alias low-pass filter *before* sampling.

The sampling theorem is commonly called Nyquist–Shannon. The same $2B$ appears in Part 3 for ideal real baseband signaling; quantization is a separate source of error.

Checkpoint — a 5 kHz tone sampled at 8 kHz

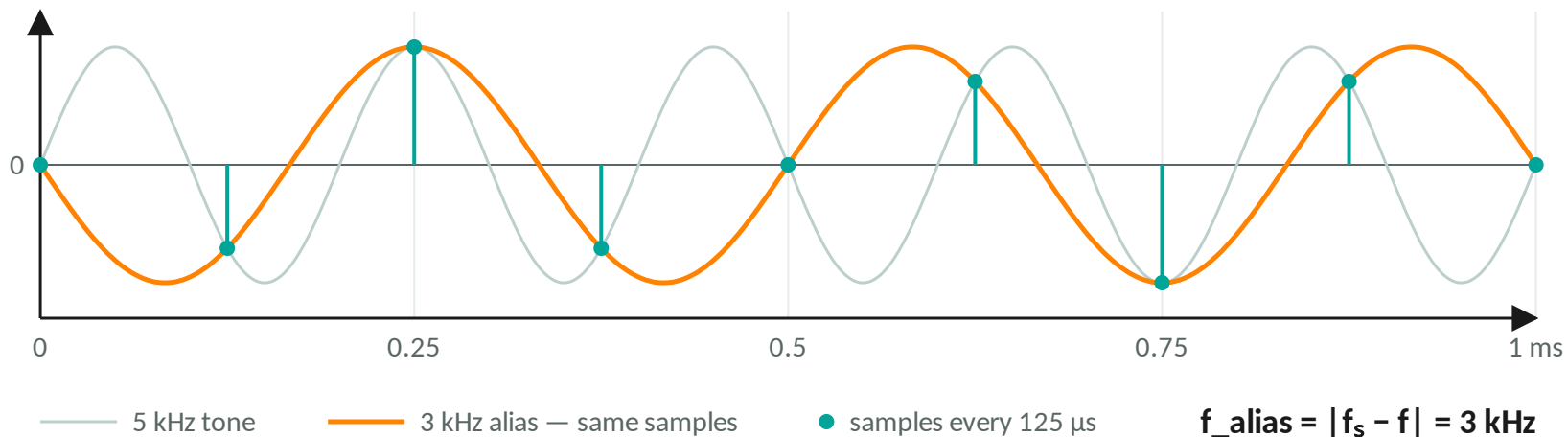
Telephone PCM samples at $f_s = 8$ kHz. A 5 kHz tone reaches the converter unfiltered. What comes out at the far end?

Predict first · discuss with a neighbour · then reveal

A 3 kHz tone.

$f_s/2 = 4$ kHz is the fold line. 5 kHz sits 1 kHz above it and lands 1 kHz below: $|f_s - f| = 8 - 5 = 3$ kHz. The samples match a 3 kHz sine with the appropriate phase.

Aliasing, seen



The grey 5 kHz tone and the orange 3 kHz tone pass through **exactly the same samples**. After sampling, the information that told them apart is gone.

Quantization: how many bits per sample?

Uniform quantization — 2^b equally spaced levels. At fixed input range, each extra bit gives about **6 dB** more signal-to-quantization-noise ratio (SQNR).

For an ideal ADC and a full-scale sine:

$$\text{SQNR}_{\text{dB}} \approx 6.02b + 1.76$$

- 8 bits → 256 levels → ≈ 50 dB
- 16 bits → 65 536 levels → ≈ 98 dB

Companding — G.711 uses A-law or μ -law: finer amplitude steps near zero, coarser steps for louder samples.

Amplitude compression before quantization and expansion after decoding improve weak-signal resolution.

It still produces **8 bits per sample**; it does not have the full-range accuracy of a higher-bit uniform quantizer.

Ideal quantization model, no clipping. [SQNR assumptions: Analog Devices MT-001](#) · [Companding: MIT PCM notes](#)

Worked example — the G.711 telephone stream

The telephone speech signal is filtered to **300–3400 Hz**.

Sampling theorem: $f_s > 2 \times 3400 = 6800$ Hz → the standard chose **8 000 Hz** (with margin for the filter).

Quantization: **8 bits** per sample, companded.

Bit rate: $8\,000 \times 8 = 64\,000$ **bit/s**.

64 kbit/s

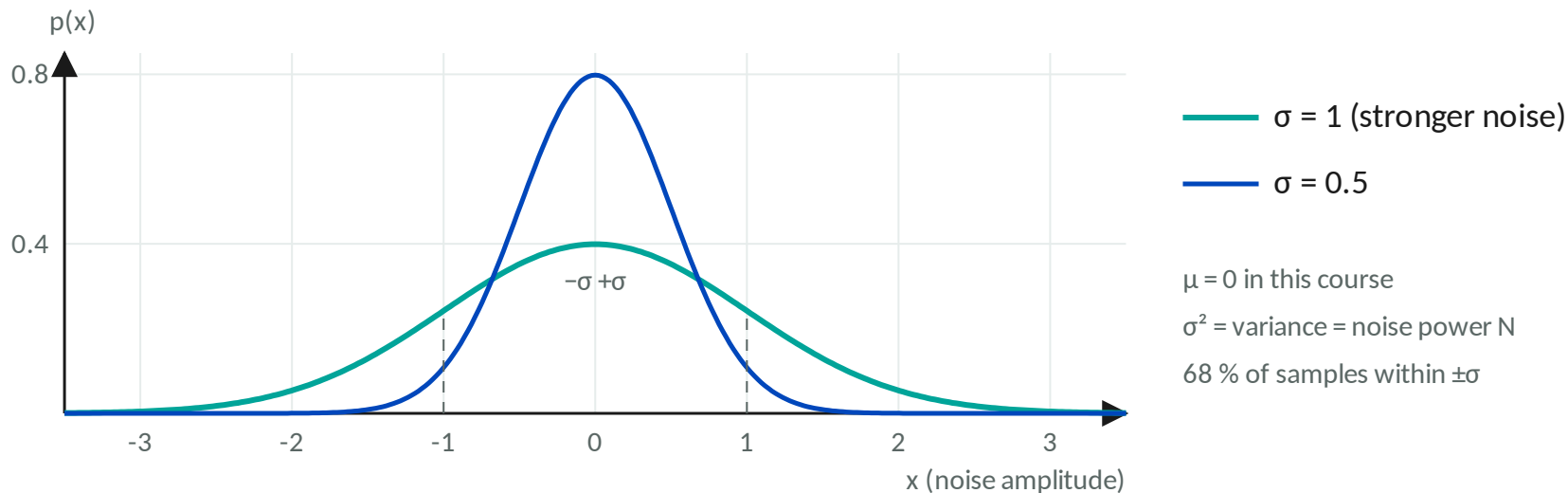
one sample every 125 μ s · one byte each · fixed bit rate

This stream is the payload of the RTP packets you will dissect in Exercise 03: 20 ms → 160 samples → 160 bytes per packet, 50 packets per second.

Part 2

Noise and the signal-to-noise ratio

Model noise as a random signal added to the useful one

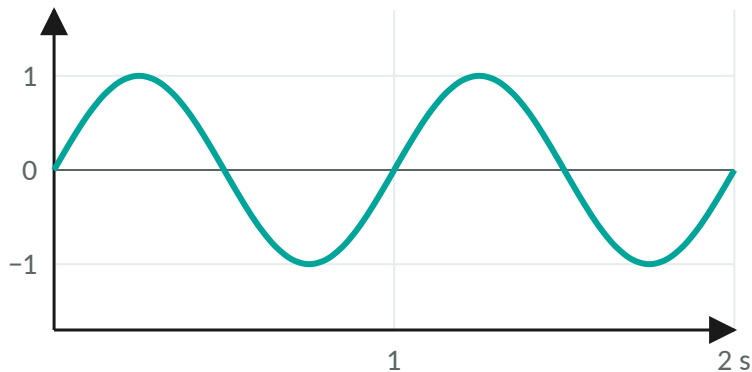


$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

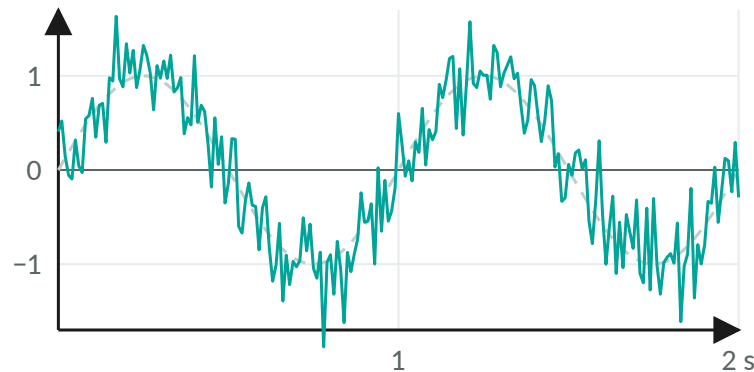
Mean μ (zero in this course) and standard deviation σ — the square root of the variance σ^2 . σ is the strength of the noise.

AWGN — the standard channel model

CLEAN SIGNAL $x(t) = \sin 2\pi ft$



$x(t) + n(t) \cdot \text{AWGN}, \sigma^2 = 0.1$



Theoretical powers: $S = 0.5 \cdot N = \sigma^2 = 0.1 \cdot S/N = 5 \approx 7 \text{ dB}$

Additive — $y = x + n$; the noise does not multiply or distort the signal, it is summed onto it.

White — flat power spectral density over the modeled band.

Gaussian — noise amplitudes follow a joint Gaussian model. Our simulation uses independent normal draws.

Signal power

For n real samples $x_1, \dots, x_n$, estimate **mean power** using

$$P = \frac{1}{n} \sum_{i=1}^n x_i^2$$

Watts if x_i are volts across $1\ \Omega$; otherwise a relative measure — and relative is all we need, because the next quantity is a ratio.

Sine, averaged over a period: $P = A^2/2$. Zero-mean noise, expected power: $P = \sigma^2$.

The previous figure has **theoretical** $S = 0.5$, $N = 0.1$. Finite-record estimates vary.

Exercise 01, Step 2 asks you to implement this before importing the reference `calc_signal_power`. Do it yourself first — it is three lines of NumPy.

Signal-to-noise ratio, linear and in decibels

$$\text{SNR} = \frac{S}{N} \quad [-]$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{S}{N} \quad \frac{S}{N} = 10^{\text{SNR}_{\text{dB}}/10}$$

Decibels are a **power** ratio on a log scale: every 10 dB is a factor of 10, every ≈ 3 dB a factor of 2.

dB	S/N
0	1
3	≈ 2
10	10
20	100
30	1 000
40	10 000

Convert to the linear ratio before the capacity formula. The formula wants S/N , not decibels.

Checkpoint — a datasheet says "SNR 20 dB"

A link has an SNR of 20 dB. What number goes into $\log_2(1 + S/N)$, and by how much would the answer be wrong if you used 20 directly?

Predict first · discuss with a neighbour · then reveal

100. $10^{20/10} = 100$, so $\log_2(101) \approx 6.66$ bit/s/Hz.

With 20 you would get $\log_2(21) \approx 4.39$ — a third of the capacity thrown away by a unit error.

Part 3

Symbols and bits — the noiseless limit

A symbol can carry more than one bit

$M = 2$ states • 1 bit per symbol



10 symbols / s

→ 10 bit/s

$M = 4$ states • 2 bits per symbol



5 symbols / s

→ 10 bit/s = 5 Bd × 2

$$R_{\text{raw}} = R_s \cdot \log_2 M \rightarrow \text{raw bit rate} = \text{symbol rate} \times \text{bits per symbol}$$

M distinct signal states label $\log_2 M$ bits per symbol when M is a power of two. Symbol rate is measured in **baud** [Bd]; raw bit rate is measured in bit/s. For binary signaling there is one raw bit per symbol.

Nyquist — ideal baseband PAM

$$R_{\max} = 2B \log_2 L$$

Pulse-amplitude modulation (PAM): a real low-pass channel of bandwidth B [Hz], L amplitude levels, no noise, and ideal pulse shaping with zero intersymbol interference.

The maximum symbol rate in this model is $2B$ baud. Each symbol labels $\log_2 L$ bits.

Example: $B = 3\,100$ Hz, $L = 16$:

$$R_{\max} = 2 \times 3100 \times 4 = 24\,800 \text{ bit/s}$$

With no noise and unlimited amplitude resolution, L and the ideal information rate are unbounded.

This $2B$ rule counts real baseband PAM symbols. Quadrature amplitude modulation (QAM) uses two coordinates per point; its constellation size is not the number of PAM levels.

One carrier frequency is not bandwidth: an ideal unmodulated tone at f_c has $B = 0$, even when f_c is large. Sending successive symbols changes the waveform and requires nonzero bandwidth.

Johnson–Nyquist noise — why N grows with B

Available thermal noise power into a matched load, in the classical regime:

$$N = kTB$$

- $k = 1.38 \times 10^{-23}$ J/K
- T — temperature [K]
- B — effective noise bandwidth [Hz]

At 290 K: **-174 dBm/Hz**.

Bandwidth	Thermal noise floor
3.1 kHz	-139 dBm
20 MHz	-101 dBm
2 GHz	-81 dBm

More bandwidth admits more noise: about **58 dB** between the first and last row.

At fixed **received signal power**, increasing bandwidth lowers SNR. Capacity then grows less than linearly.

Checkpoint — why not a million levels?

Nyquist says more levels means more bits per symbol at the same bandwidth. What stops us from using $L = 10^6$ and getting 20 bits per symbol?

Predict first • discuss with a neighbour • then reveal

Noise. In a fixed amplitude range, adding levels shrinks their spacing. When that spacing is comparable to σ , decisions between neighboring levels become error-prone.

Closer signal points are harder to distinguish in noise. **Shannon accounts for coding across many symbols** to find the ultimate reliable information rate; there is no universal level-count formula based on SNR alone.

Part 4

The Shannon-Hartley theorem

Capacity of a noisy channel

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

C — channel capacity [bit/s]

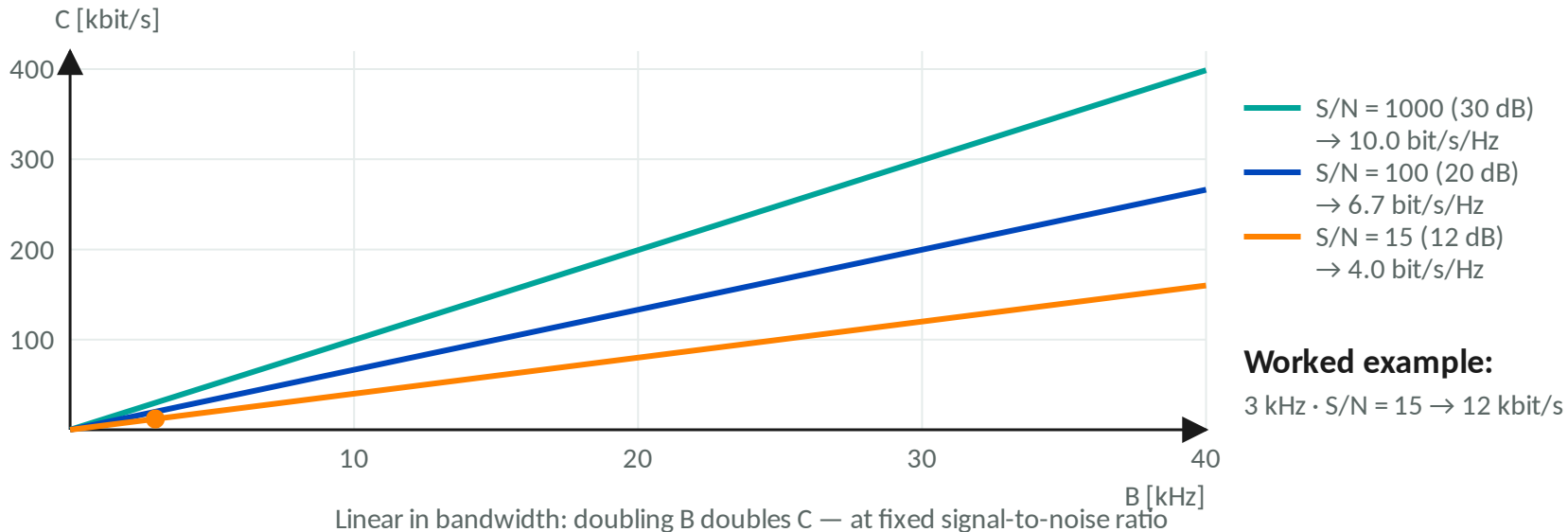
B — bandwidth [Hz]

S/N — **linear** signal-to-noise ratio
[-]

With bandwidth B , fixed average **received** signal power S , and independent AWGN of power N over that band: any information rate below C can achieve **arbitrarily small error probability** with sufficiently long codes.

Above C , arbitrarily reliable communication is impossible under this model. Approaching C generally costs coding complexity and delay — a practical concern for voice.

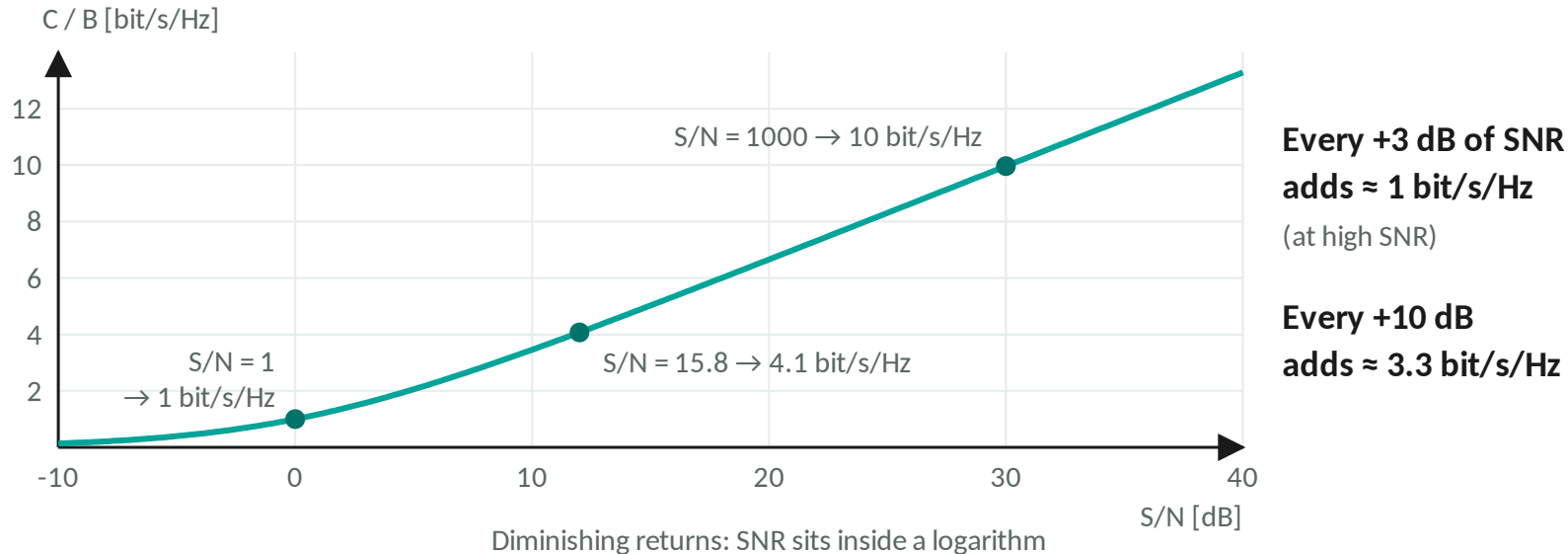
Knob 1 — bandwidth is linear



At fixed S/N , **doubling B doubles C** . Wider channels are one way to increase rate; the condition **fixed SNR** matters.

The catch: thermal noise is $N = kTB$ (Johnson-Nyquist), so if the signal power stays fixed, a wider channel also lowers S/N . "Fixed SNR" must be checked, not assumed.

Knob 2 — SNR is logarithmic



At high SNR, doubling S/N adds approximately **1 bit/s/Hz**; multiplying it by ten adds approximately **3.3 bit/s/Hz**. With unchanged propagation and noise, more transmit power has diminishing returns.

Worked example

A channel with $B = 3\,000$ Hz and $S/N = 15$.

$$C = 3000 \cdot \log_2(1 + 15) = 3000 \cdot \log_2 16$$

$$C = 3000 \cdot 4 = 12\,000 \text{ bit/s}$$

12 kbit/s

for 3 kHz of bandwidth at 11.8 dB

Units check: B is bandwidth [Hz]; C is an information rate [bit/s].

Their ratio $C/B = \log_2(1 + S/N)$ is **spectral efficiency**, measured in bit/s/Hz.

Checkpoint — 64 kbit/s through a 33 kbit/s channel?

For an analog voiceband channel, $B = 3.1$ kHz and $S/N = 1585$ give $C \approx 33$ kbit/s. G.711 produces 64 kbit/s. Is the capacity bound being violated?

Predict first · discuss with a neighbour · then reveal

No — these rates describe different channels.

The 3.1 kHz band describes the analog voiceband circuit. The 64 kbit/s PCM representation travels over a **digital transport link** with its own bandwidth, noise, and capacity.

The 33 kbit/s value is a result for the stated assumptions, not a universal limit for every telephone connection.

Capacity, coded bits, and useful data

Physical-channel quantities

Capacity C : the reliable information-rate limit for the specified channel model.

Raw coded bit rate: includes error-correction redundancy. It can exceed C .

Net information rate: $R_{\text{info}} = r R_{\text{raw}}$, where r is information bits / coded bits. Reliable operation must respect C .

Measured network quantities

Throughput: successful transfer rate at a stated layer, such as IP. Contention, loss, and retransmissions affect it.

Goodput: useful application payload delivered per second, excluding protocol overhead and duplicates.

Codec output is usually application payload.

QoS manages **rate, delay, jitter, loss, and fairness** within the available resources.

From 64 kbit/s of speech to 80 kbit/s of IP traffic

G.711, 20 ms packetization

- 160 bytes of speech per packet
- RTP 12 B + UDP 8 B + IPv4 20 B
- 50 packets/s, in each direction

$$R_{\text{IP}} = (160 + 40) \times 8 \times 50$$

80 kbit/s at IP; 64 kbit/s of payload.

What if packets cover 40 ms?

320 bytes of speech, 25 packets/s:

$$R_{\text{IP}} = (320 + 40) \times 8 \times 25$$

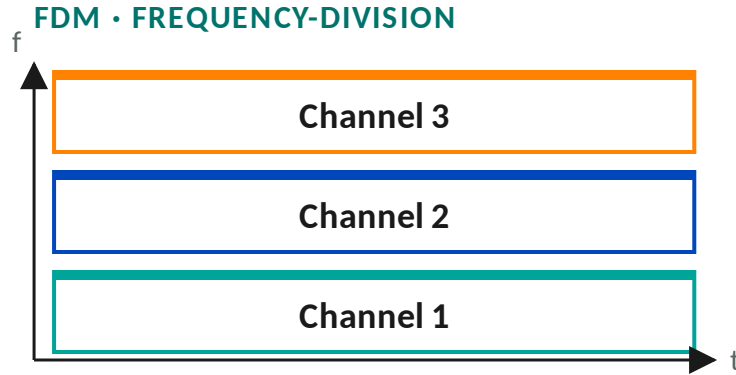
72 kbit/s at IP; still 64 kbit/s of payload.

Less overhead per second, but more packetization delay and more speech affected by each lost packet.

No IPv4 options, no RTP extensions, no link-layer overhead; continuous speech transmission. Same calculation as Exercise 03.

ČTÚ note: the Czech Telecommunication Office (Český telekomunikační úřad) requires access-line speed to always be measured at **OSI layer 4** — transport-layer throughput, not the raw physical or IP-layer rate.

Sharing one channel — FDM and TDM

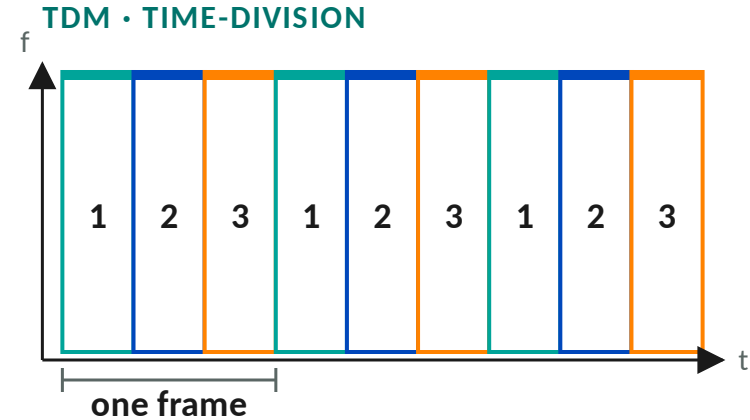


Every channel transmits all the time, in its own slice of bandwidth.

□ Channel 1

□ Channel 2

□ Channel 3



Every channel gets the full bandwidth, once per frame.

Frequency-division multiplexing (FDM) splits the bandwidth into sub-bands, one per channel. Analog radio and TV; guard bands stop them bleeding into each other.

Time-division multiplexing (TDM) splits time into slots, one per channel, repeating every frame. Telephony trunks (T1/E1), and the scheduling inside Wi-Fi and LTE.

MIMO — a third axis, space, to divide a channel by

Multiple-input multiple-output (MIMO) uses several transmit and receive antennas.

Different spatial paths can let the receiver separate several streams **in the same frequency band**.

With n_t transmit and n_r receive antennas, there are at most $\min(n_t, n_r)$ spatial streams.

Antennas do not guarantee streams

The channel must distinguish the transmitted signals well enough. Similar paths may provide only one useful mode.

Available power must be allocated across streams; each has its own SNR.

More antennas can also improve reliability or received signal strength.

Part 5

Measuring information — Hartley and Shannon

How much information is in one letter?



Each question halves the set: $26 \rightarrow 13 \rightarrow 7 \rightarrow 4 \rightarrow 2 \rightarrow 1$

Five suffice in the worst case; an optimal separate-letter tree averages 4.769 questions

$\log_2 26 = 4.700$ bits of information

Information $\neq$ integer question count

Someone picks one of **26 equally likely letters**. You may ask yes/no questions. How many do you need in the worst case?

Hartley's measure — 1928

26 equally likely letters:

$$I_{\text{Hartley}} = \log_2 26 \approx 4.700 \text{ bits}$$

Five yes/no questions suffice in the worst case.

An optimal separate-letter question tree averages **4.769 questions**, not 4.700.

Blocks of n independent uniform letters:

There are 26^n possible messages. A fixed-length binary block code needs

$$\lceil n \log_2 26 \rceil \text{ bits.}$$

As blocks grow, bits per letter approach **4.700**.

For all s^n possible messages: $I_{\text{Hartley}} = n \log_2 s$. Base 2 gives bits; base e gives nats; base 10 gives hartleys.

Unequal probabilities change the questions

Equal probabilities

Divide the **number of candidates** roughly in half.

For 26 uniform letters, a split of 13 versus 13 is balanced.

Unequal probabilities

Divide the **remaining probability mass** roughly in half.

Common outcomes can receive shorter descriptions.
Guessing the most frequent letter first is not generally the best yes/no question.

Shannon's measure accounts for how likely each outcome is. We will calculate it for a four-outcome source and build a code.

Self-information of one outcome

$$I(x_i) = \log_2 \frac{1}{p_i} = -\log_2 p_i \quad [\text{bit}]$$

- $p = 1 \rightarrow 0$ bits. A certain outcome tells you nothing.
- $p = 1/2 \rightarrow 1$ bit. A fair coin.
- $p = 1/26 \rightarrow 4.7$ bits. Hartley's uniform letter.
- If $p = 1/4 \rightarrow 2$ bits
- If $p = 1/8 \rightarrow 3$ bits

The rarer the outcome, the more it tells you.
Information is surprise, measured in bits.

Information measures **uncertainty under a probability model**. It does not measure a message's importance, usefulness, or truth.

Shannon entropy — the average information per symbol

$$H = \sum_i p_i \log_2 \frac{1}{p_i} = - \sum_i p_i \log_2 p_i \quad [\text{bit/symbol}]$$

One symbol repeated — $p = 1$ for it, 0 for all others:
 $H = 0$. With the source and framing known, each next symbol is certain.

All N symbols equally likely — $p_i = 1/N$: $H = \log_2 N$.
Hartley's value, and the **maximum** possible for N symbols.

$$0 \leq H \leq \log_2 N$$

For a fixed alphabet of N symbols, the maximum occurs only at the uniform distribution. Use $0 \log_2 0 := 0$.

A source and a code we can calculate

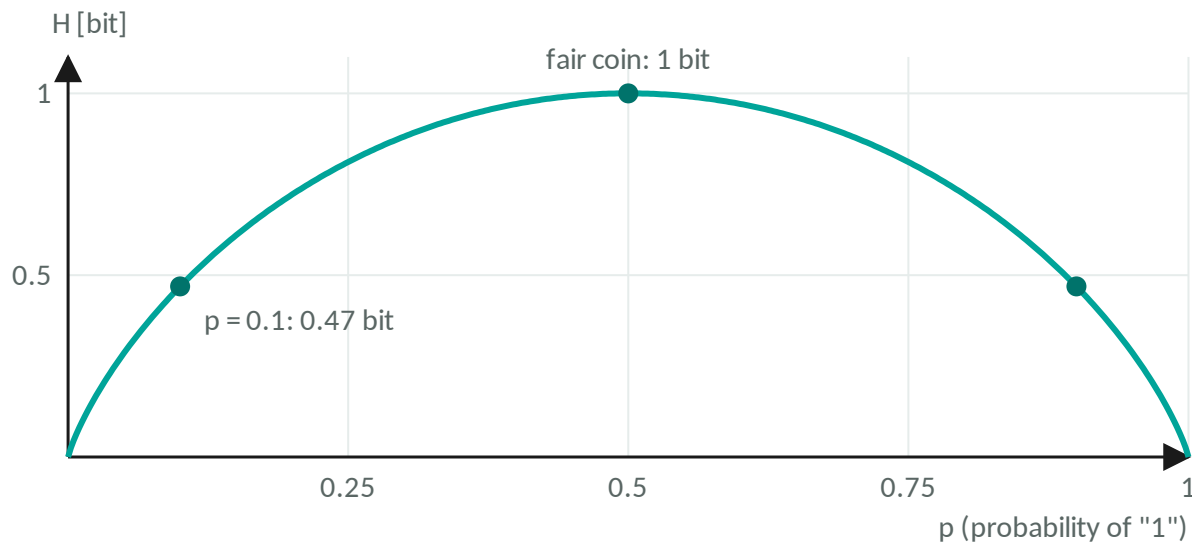
Outcome	Probability	Information $-\log_2 p$	Code	Length
A	$1/2$	1 bit	0	1
B	$1/4$	2 bits	10	2
C	$1/8$	3 bits	110	3
D	$1/8$	3 bits	111	3

$$H = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75 \text{ bit/symbol}$$

Average code length is also **1.75 bits/symbol**, versus 2 for a fixed-length code.

Decode: 010110111 $\rightarrow A \cdot B \cdot C \cdot D$

The binary source



Certain outcome

$p = 0$ or $1 \rightarrow H = 0$

Most uncertain

$p = 0.5 \rightarrow H = 1$ bit

Uniform is the maximum:

$H \leq \log_2 N$, equal iff uniform

$$H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

Independent biased tosses with $p = 0.1$ have $H \approx 0.469$ bit per toss. With a known model and suitable block coding, 1 000 tosses need **about 469–470 bits on average**, not for every possible sequence.

Same symbol counts, different predictability

Independent fair bits

```
011010001101...
```

Each new bit is equally likely to be 0 or 1, even after seeing the past.

Entropy rate: 1 bit per symbol.

A known alternating rule

```
010101010101...
```

The counts are also half 0, half 1. Once the starting bit is known, every next bit is determined.

Entropy rate: 0 bits per symbol.

Single-symbol frequencies miss dependencies. For a source with memory, compression is limited by its **entropy rate**: the uncertainty per symbol that remains after accounting for the past.

Three different jobs called encoding

Stage	What it does	Example
PCM encoding	Labels quantized samples with bit patterns	G.711: 8 bits per sample
Lossless source coding	Removes statistical redundancy	Shorter codes for common outcomes
Channel coding	Adds controlled redundancy for error protection	Extra coded bits to recover damaged data

Lossy source coding can use fewer bits by accepting distortion. G.711 quantization and speech codecs such as Opus do not preserve the original analog waveform exactly.

The two theorems, and where QoS lives

Source coding

For independent symbols, lossless coding can **approach H bits per symbol on average** with long blocks, but cannot beat that average limit. For dependent sources, use the entropy rate.

How many bits does the message need?

Channel coding

Under the stated AWGN constraints, any information rate below C can achieve arbitrarily small error with sufficiently long codes. No rate above C can do so.

How many bits can the channel carry?

A codec produces a stream; packets and protection add overhead; a link serves the traffic. **Queues, scheduling, admission, and congestion control** help meet rate, delay, jitter, and loss requirements.

Summary — five formulas to keep

Sampling — ideal low-pass, unquantized samples

$$f_s > 2f_{\max}$$

Signal-to-noise ratio — convert to linear first

$$S/N = 10^{\text{SNR}_{\text{dB}}/10}$$

Nyquist — ideal real baseband PAM, L levels

$$R_{\max} = 2B \log_2 L$$

Shannon-Hartley — AWGN, specified power and band

$$C = B \log_2(1 + S/N)$$

Entropy — average surprise for one source symbol

$$H = - \sum_i p_i \log_2 p_i$$

Capacity is linear in bandwidth **at fixed SNR**. A binary symbol has 1 bit of entropy only when its two outcomes are equally likely; dependencies require entropy rate.

Exit questions — use the ideas

1. Does 0 dB SNR mean zero channel capacity?
2. Does doubling bandwidth double capacity at fixed received signal power?
3. Can a binary source have less than one bit of entropy per symbol?
4. Why does a 64 kbit/s G.711 payload produce 80 kbit/s of IPv4 traffic?

Explain the assumptions and units, not just the formula.

Exercise 01 — what you will do

1. **Model a signal with noise** — a harmonic signal, Gaussian noise with a σ slider, watch the SNR move.
2. **Implement the channel metrics** — signal power, SNR, Shannon capacity — *before* importing the reference versions from `lib.core`.
3. **Noise on audio** — add AWGN to a WAV file and listen to what 20 dB, 10 dB and 0 dB sound like.
4. **Noise on an image** — the same, seen.

Then the README questions:

- SNR units and the linear ratio
- MIMO and why antennas $\neq$ streams
- 64 kbit/s vs the 33 kbit/s telephone channel
- the 5 kHz tone at $f_s = 8$ kHz

```
1  ssh <lab-server>
2  cd qos-01 && uv sync
3  uv run jupyter lab --ip 0.0.0.0
```

References — foundations

- C. E. Shannon, "[A Mathematical Theory of Communication](#)," 1948.
- C. E. Shannon, "Communication in the Presence of Noise," *Proc. IRE*, 37(1), 10–21, 1949.
- C. E. Shannon, "[Prediction and Entropy of Printed English](#)," 1951.
- R. V. L. Hartley, "Transmission of Information," *Bell System Technical Journal*, 7, 535–563, 1928.
- H. Nyquist, "Certain Topics in Telegraph Transmission Theory," *Trans. AIEE*, 47, 617–644, 1928.
- H. Nyquist, "[Thermal Agitation of Electric Charge in Conductors](#)," 1928.

References — explanations and applications

- [MIT: PCM, sampling, and companding](#); [Huffman coding](#); [source-coding theory](#).
- Analog Devices, [MT-001: quantization SNR and its assumptions](#).
- D. Tse and P. Viswanath, [Fundamentals of Wireless Communication](#), chapters 7–8.
- ITU-T [G.711](#), [V.90](#), and [V.92](#); [RFC 6716: Opus](#).
- [ETSI TS 138 104 v19.2.0](#), NR channel bandwidths.

Thank you for your attention

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Extras

Optional history, comparisons, and mathematical detail

Extras — the course in thirteen lectures

1. Channel capacity, information theory, entropy ← today
2. Kendall's classification of queueing systems; basic models
3. QoS vs QoE; approaches to achieving QoS
4. Models of network traffic; qualitative parameters in IP networks
5. Evaluating the quality of multimedia services I
6. Evaluating the quality of multimedia services II
7. Quality of Internet access services (ČTÚ, invited lecture)
8. Network modelling in software tools
9. The SDN concept and its use
10. Quality of service in SDN networks I
11. Quality of service in SDN networks II
12. Quality of service in SDN networks III
13. QoS in quantum communication infrastructure

Lab exercises (`qos-01` ... `qos-05`) run on the laboratory servers over SSH. Each lecture has a matching README with the theory and a notebook with the tasks.

Extras — signal classification

Criterion	Classes	Example
Amplitude	<i>analog</i> — any value in a continuous range <i>digital</i> — a finite set of values, usually binary codes	microphone output computer data
Time	<i>continuous-time</i> — defined at every instant <i>discrete-time</i> — defined only at sampling instants	voltage on a wire samples in a WAV file
Repetition	<i>periodic</i> — repeats with period T <i>aperiodic</i> — has no repeating period	sine wave a single pulse
Predictability	<i>deterministic</i> — given exactly by a function <i>random</i> — described only statistically	$\sin(2\pi ft)$ thermal noise

The axes are independent: a WAV file is digital *and* discrete-time; noise is analog, continuous-time *and* random.

Extras — why we go digital

Regeneration

Analog repeaters also amplify noise. A digital regenerator decides a symbol and sends a clean waveform — correctly if the decision was correct.

Processing

Once it is numbers, it can be compressed, encrypted, mixed, stored and routed by a computer.

Multiplexing

Bits from many sources interleave on one link. Packets are this idea taken to its conclusion.

For an analog source, conversion adds two concerns: **aliasing** and **quantization error**. Part 1 is how to keep both under control.

Extras — modulation states and bit rates

$$R_{\text{raw}} = R_s \log_2 M$$

R_{raw} — raw coded bit rate [bit/s] · R_s — symbol rate [Bd] · M — signal states

Scheme	Distinct states M	Bits per symbol	At 1 000 Bd
Binary PAM	2 amplitudes	1	1 000 bit/s
PAM-4	4 amplitudes	2	2 000 bit/s
QPSK	4 phases	2	2 000 bit/s
16-QAM	16 constellation points	4	4 000 bit/s

With channel-code rate r , the information bit rate is $R_{\text{info}} = r R_{\text{raw}}$ before framing and protocol overhead.

Extras — Nyquist's contributions

Year	Paper	Contribution
1924	<i>Certain Factors Affecting Telegraph Speed</i>	Telegraph speed, bandwidth, and signal levels
1928	<i>Certain Topics in Telegraph Transmission Theory</i>	Ideal pulse transmission and the $2B$ rate
1928	<i>Thermal Agitation of Electric Charge in Conductors</i>	Thermal-noise theory
1932	<i>Regeneration Theory</i>	Feedback stability criterion

Born in Sweden in 1889; PhD at Yale in 1917. Worked in the Bell System until 1954 and received 138 US patents.

[Contemporary biography: Bell Laboratories Record, November 1960](#) · [Shannon's 1948 introduction cites Nyquist and Hartley](#)

Extras — bandwidth conventions

Real baseband PAM

L amplitude levels, symbol rate R_s , roll-off α :

$$B = \frac{(1 + \alpha)R_s}{2}$$

At $\alpha = 0$: $R_{\max} = 2B \log_2 L$.

Passband QAM

M two-dimensional constellation points:

$$B = (1 + \alpha)R_s$$

At $\alpha = 0$: $R_{\text{raw}} = B \log_2 M$.

B is occupied bandwidth [Hz]; R_s is symbols/s. Both formulas assume ideal raised-cosine pulse shaping. Noise and coding determine the reliable information rate.

Extras — a carrier frequency is not bandwidth

$$x(t) = A \cos(2\pi f_c t + \phi)$$

A — amplitude; f_c — carrier frequency [Hz]; ϕ — phase [rad].

A pure tone

With A , f_c , and ϕ fixed forever, the ideal tone occupies **one frequency**: $B = 0$.

A 1 GHz carrier has a billion cycles per second, but those predictable cycles are not a billion independent symbols.

Sending new choices

Successive symbols must change the waveform.

Amplitude changes introduce **sidebands** around the carrier.

A 1 MHz-wide band centered at 1 GHz still has only **1 MHz of bandwidth**.

Extras — where AM sidebands come from

Vary a carrier's amplitude with a message tone of frequency f_m :

$$x(t) = A_c[1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

A_c — carrier amplitude; μ — modulation depth; $f_c > f_m > 0$ — frequencies [Hz].

Use $\cos a \cos b = \frac{1}{2} \cos(a + b) + \frac{1}{2} \cos(a - b)$:

$$\begin{aligned} x(t) &= A_c \cos(2\pi f_c t) \\ &\quad + \frac{\mu A_c}{2} \cos(2\pi(f_c + f_m)t) \\ &\quad + \frac{\mu A_c}{2} \cos(2\pi(f_c - f_m)t). \end{aligned}$$

Carrier: f_c · **Upper sideband:** $f_c + f_m$ · **Lower sideband:** $f_c - f_m$

Extras — calculate the sideband frequencies

Carrier $f_c = 1$ MHz; message $f_m = 2$ kHz; $A_c = 1$ V; modulation depth $\mu = 0.5$.

Component	Frequency	Peak amplitude
Lower sideband	$f_c - f_m = 998$ kHz	$\mu A_c / 2 = 0.25$ V
Carrier	$f_c = 1\,000$ kHz	$A_c = 1$ V
Upper sideband	$f_c + f_m = 1\,002$ kHz	$\mu A_c / 2 = 0.25$ V

The span from the lower to the upper sideband is **4 kHz**, even though the carrier is at 1 MHz.

For a real message occupying baseband frequencies up to W , double-sideband AM occupies $f_c - W$ to $f_c + W$: **passband width** $2W$, assuming $f_c > W$.

Extras — sidebands of a digital PAM stream

A symbol stream has many frequency components. **Symbol rate and pulse shape** determine its spectral extent.

Ideal raised-cosine pulse shaping

For symbol rate R_s [Bd] and roll-off $0 \leq \alpha \leq 1$:

$$W = \frac{(1 + \alpha)R_s}{2}$$

W is real baseband bandwidth [Hz]. Multiplying by a carrier shifts the spectrum to $f_c - W$ through $f_c + W$.

Passband width: $2W = (1 + \alpha)R_s$.

Example

$R_s = 1\,000$ Bd, $\alpha = 0.25$:

$$W = \frac{1.25 \times 1000}{2} = 625 \text{ Hz}$$

Around a 1 MHz carrier:

999.375–1 000.625 kHz

Total passband width: **1.25 kHz**.

Abrupt rectangular pulses have spectral tails extending indefinitely. Practical bandwidth then needs a convention, such as first-null bandwidth or a specified power fraction.

Extras — the wideband limit

At fixed **received** signal power S and thermal-noise density kT :

$$C(B) = B \log_2 \left(1 + \frac{S}{kTB} \right) \xrightarrow{B \rightarrow \infty} \frac{S}{kT \ln 2}$$

- C — capacity [bit/s]; B — effective bandwidth [Hz]
- S — average received signal power [W]
- kT — noise power spectral density [W/Hz]

At low SNR, $\ln(1 + x) \approx x$: increasing bandwidth spreads the fixed signal power over more noise. Capacity grows toward a **finite limit**.

Extras — compare illustrative channels

Model / configuration	Bandwidth B	Assumed linear S/N	AWGN capacity
Analog voiceband example	3.1 kHz	1 585	33.0 kbit/s
One idealized DSL tone	4.3125 kHz	1 000	43.0 kbit/s
Idealized 20 MHz radio	20 MHz	100	133.2 Mbit/s
NR FR2-1 example, 28 GHz	400 MHz	6.3	1.147 Gbit/s

SNRs are hypothetical. These are single-channel AWGN calculations, not promised service rates. DSL tones have different SNRs; real radios add coding constraints, guard intervals, overhead, and possibly spatial streams.

400 MHz is a component-carrier option in FR2-1; aggregation is separate. [ETSI TS 138 104, §5.3.5](#)

Extras — MIMO capacity by spatial mode

A fixed MIMO channel can be decomposed into m parallel spatial modes:

$$C = B \sum_{i=1}^m \log_2(1 + \text{SNR}_i), \quad m \leq \min(n_t, n_r)$$

Here B is bandwidth [Hz]; SNR_i is the linear SNR of mode i after its power allocation. Capacity uses the optimal allocation under the specified power constraint.

For equal per-mode SNR γ , this becomes $mB \log_2(1 + \gamma)$. **Keeping total transmit power fixed does not keep every mode's SNR fixed.**

Mode gains depend on geometry, scattering, and polarization. Line of sight alone does not imply rank one. [Tse and Viswanath, chapters 7–8](#)

Extras — entropy estimates for English

Model	Approximate bits / letter
26 equally likely letters	4.70
Unequal letter frequencies	4.14
Also condition on the previous letter	3.56

Shannon's long-context guessing experiment used **27 symbols, including space**, and found experimental bounds around **0.6–1.3 bits/symbol** for its longest context.

A model-dependent estimate

Structure makes the next symbol more predictable. These numbers depend on the text, alphabet, and model.

Plain ASCII stored one byte per character uses 8 bits. A lossless compressor exploits statistical structure to reduce the average length.

Entropy rate is the asymptotic average limit, not a promised size for every file.